(3) Derivative of the logarithmic function

The natural logarithm of x is denoted by $\log_e x$ or $\log x$ or $ln\ x$

$$\text{Let } y = f(x) = \log x$$

$$\text{Now } f(x+\Delta x) = \log(x+\Delta x) \text{ and}$$

$$f(x+\Delta x) - f(x) = \log(x+\Delta x) - \log x$$

$$= \log\left(\frac{x+\Delta x}{x}\right).$$

$$= \log\left(1+\frac{\Delta x}{x}\right).$$

$$\frac{f(x+\Delta x)-f(x)}{\Delta x} = \frac{\log\left(1+\frac{\Delta x}{x}\right)}{\Delta x}.$$

We know that $\lim\limits_{\alpha\to 0}\frac{\log(1+\alpha k)}{\alpha} = k\lim\limits_{\alpha\to 0}\frac{\log(1+k\alpha)}{k\alpha} = k$

$$\text{Therefore, } \lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = \lim_{\Delta x\to 0}\frac{\log\left(1+\frac{\Delta x}{x}\right)}{\Delta x} = \frac{1}{x}.$$

That is, $\boxed{\frac{d}{dx}(\log x) = \frac{1}{x}}$.

Corollary 10.3

$$\text{If } y = f(x) = \log_a x \text{ then } f'(x) = \frac{1}{(\log a)x}.$$

We have $f(x) = \log_a x = \log_a e \times \log_e x = (\log_a e)\log x$

$$\text{Therefore, } \frac{d}{dx}\big(f(x)\big) = \frac{d}{dx}(\log_a e \times \log x)$$

$$= (\log_a e)\frac{d}{dx}(\log x) \qquad \text{(by constant multiple rule)}$$

$$= \log_a e.\frac{1}{x} \text{ (or) } \frac{1}{(\log a)x}.$$

(4) Derivative of the exponential function

$$\text{Let } y = a^x$$

$$\text{Then } f(x+\Delta x) - f(x) = a^{x+\Delta x} - a^x$$

$$= a^x(a^{\Delta x}-1) \text{ and}$$

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